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Topic 6 — Groups in the Periodic Table

Edexcel GCSE (9-1) Chemistry (1CH0) — Paper 2

This topic looks closely at three groups of the periodic table: **Group 1** (the alkali metals), **Group 7** (the halogens) and **Group 0** (the noble gases). Elements in the same group have the **same number of electrons in their outer shell**, so they react in similar ways — and as you go down a group there are clear **trends** in physical properties and reactivity. The key skill examiners test is using a pattern to **predict** the properties of an element you have not studied (e.g. rubidium, fluorine, astatine, krypton) and **explaining** trends in terms of **electronic configurations**.

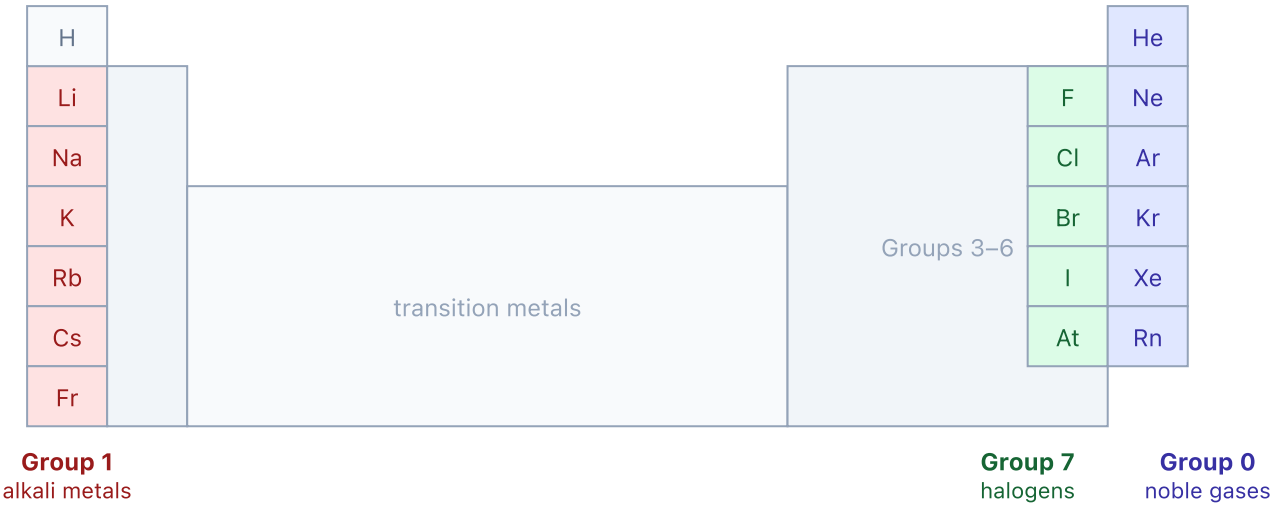
Content marked **(HT)** is Higher Tier only. Topic 6 is common to GCSE Chemistry and Combined Science — there is no chemistry-only content and no core practical in this topic.

6.1 Classifying Groups 1, 7 and 0

> The **group number** (for Groups 1–7) equals the **number of electrons in the outer shell**. Elements are classified as **alkali metals** (Group 1), **halogens** (Group 7) or **noble gases** (Group 0) because of their **position** in the periodic table, which reflects their outer-shell electron arrangement.

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Groups 1, 7 and 0 in the periodic table



Group	Name	Outer-shell electrons	Typical behaviour
1	Alkali metals	1	Lose 1 electron to form 1+ ions (e.g. Na^+); very reactive metals
7	Halogens	7	Gain 1 electron to form 1– ions (halide ions, e.g. Cl^-); reactive non-metals; exist as diatomic molecules (Cl_2)
0	Noble gases	Full outer shell (2 for He, 8 for the others)	Very unreactive (inert); exist as single atoms (monatomic)

Exam Tip: Helium has only **2** outer electrons, yet it is in Group 0 — because its outer shell (the first shell) is **full**. If asked why helium is placed in Group 0 rather than Group 2, say it has a full outer shell and is unreactive like the other noble gases.

6.2 Group 1 — The Alkali Metals

Physical properties

The alkali metals are unusual metals:

- They are **soft** — they can be cut with a knife (softness increases down the group).
- They have **relatively low melting points** (decreasing down the group).
- They have **low densities** — lithium, sodium and potassium float on water.
- They are shiny when freshly cut but quickly **tarnish** as they react with oxygen in the air, so they are stored under **oil**.

Element	Electronic configuration	Melting point / °C	Density / g cm^{-3}
Lithium, Li	2.1	181	0.53
Sodium, Na	2.8.1	98	0.97
Potassium, K	2.8.8.1	63	0.86
Rubidium, Rb	2.8.18.8.1	39	1.53
Caesium, Cs	2.8.18.18.8.1	28	1.93

Melting point decreases steadily down the group. Density generally increases (potassium is an exception).

Reactions with water

All alkali metals react with cold water to produce a **metal hydroxide** and **hydrogen**. The hydroxide dissolves to form an **alkaline** solution (pH above 7) — which is why they are called alkali metals.

alkali metal + water → metal hydroxide + hydrogen

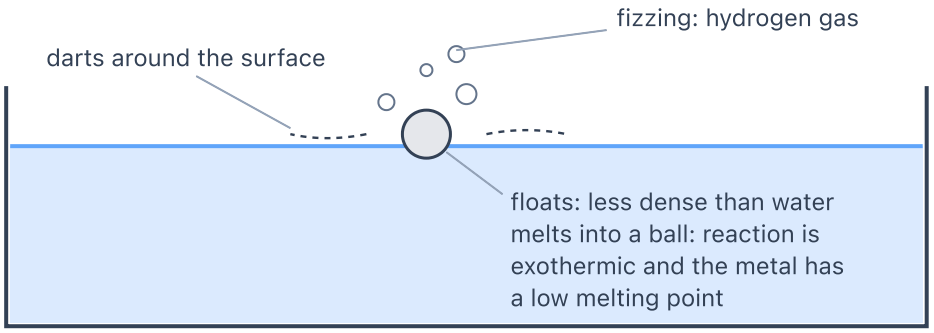
$$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$$

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$$

$$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$$

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Alkali metal reacting with water in a trough



add universal indicator: turns purple (alkaline hydroxide forms)

Metal	Observations with cold water
Lithium	Floats; fizzes steadily; moves slowly around the surface; gradually disappears. Does not melt.
Sodium	Floats; fizzes rapidly; **melts into a silvery ball**; moves quickly around the surface; disappears quickly. The hydrogen may ignite with an orange flame if the sodium is held still.
Potassium	Floats; melts; fizzes very violently; the hydrogen **ignites spontaneously with a lilac flame**; may spit and crackle; disappears very quickly.

Exam Tip: "Describe" questions want **observations** — things you would see or hear (fizzes, floats, melts, moves, lilac flame, gets smaller). "Hydrogen is produced" is a conclusion, not an observation; "bubbles/fizzing" is the observation. Never say "a gas is produced" when you mean "effervescence".

The pattern in reactivity — and predictions

Reactivity with water **increases down Group 1**: lithium < sodium < potassium. Following the pattern:

- **Rubidium** will react even more violently than potassium — it may explode; it will melt instantly and the hydrogen will ignite.
- **Caesium** reacts explosively with water.

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- All will still form a **metal hydroxide** (RbOH, CsOH) and **hydrogen**.

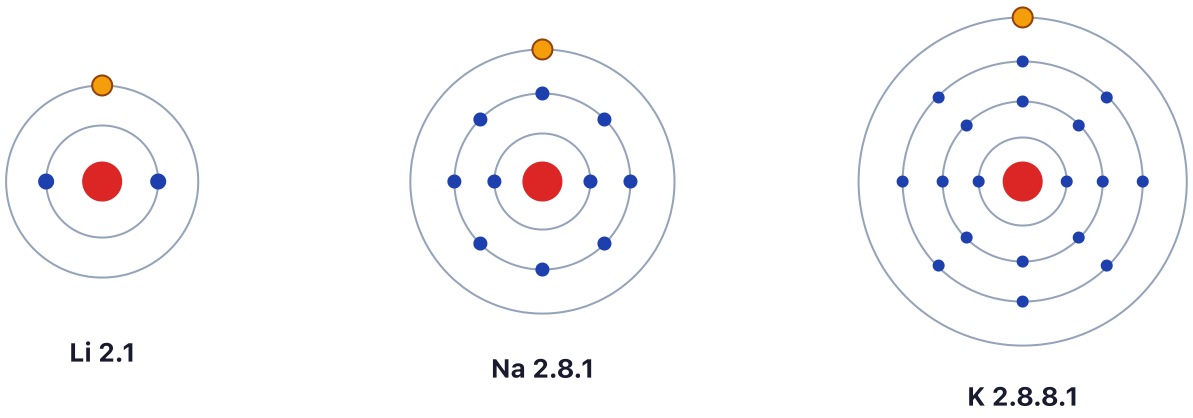
> **Worked Example:** Predict the equation and the observations for the reaction of rubidium with water.
>
> Equation (same pattern as Li, Na, K): **$2Rb(s) + 2H_2O(l) \rightarrow 2RbOH(aq) + H_2(g)$**
> Observations: rubidium is below potassium so it is more reactive; it would react **more violently** than potassium — immediate melting, very rapid fizzing, the hydrogen ignites and the reaction may be **explosive**. The solution formed is strongly **alkaline**.

Explaining the trend in terms of electronic configuration

When an alkali metal reacts, its atoms **lose their single outer electron** to form a 1+ ion:

$$Na \rightarrow Na^+ + e^-$$

Outer electron further from the nucleus down Group 1



Orange = the single outer electron that is lost. More shells → weaker attraction → lost more easily.

Going down the group:

1. The atoms have **more electron shells**, so the atoms get **larger**.
2. The outer electron is **further from the nucleus** (and more shielded by inner shells).
3. So the **attraction** between the positive nucleus and the outer electron is **weaker**.
4. The outer electron is therefore **lost more easily**, so the metal is **more reactive**.

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Exam Tip: This is a classic 3–4 mark explanation. Work in a chain: **more shells → outer electron further from nucleus → weaker attraction → electron lost more easily → more reactive**. Do not say "the nucleus has less charge" — the nuclear charge actually increases down the group.

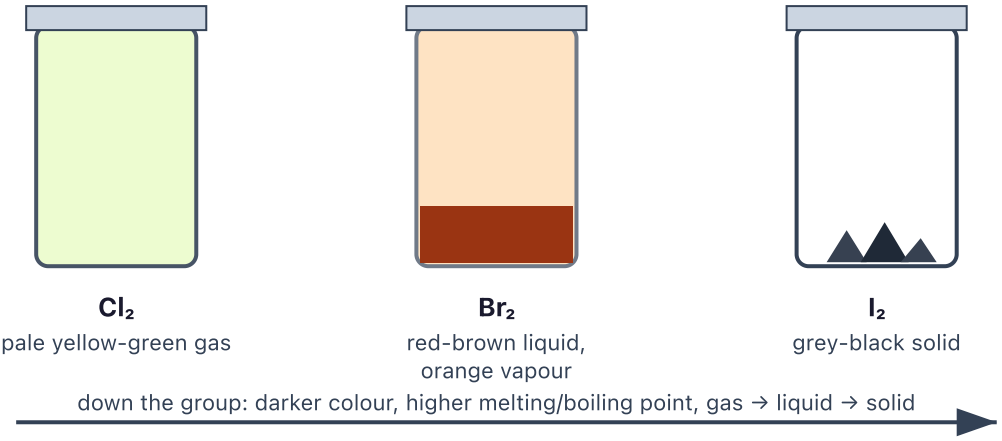
6.3 Group 7 — The Halogens

The halogens are non-metals that exist as **diatomic molecules** (F_2, Cl_2, Br_2, I_2) held together by a single covalent bond. They are toxic and have strong smells.

Colours and physical states at room temperature

Halogen	Formula	State at room temperature	Colour	Melting point / °C	Boiling point / °C
Fluorine	F_2	Gas	Pale yellow	−220	−188
Chlorine	Cl_2	**Gas**	**Pale yellow-green**	−101	−34
Bromine	Br_2	**Liquid**	**Red-brown** (gives off an orange-brown vapour)	−7	59
Iodine	I_2	**Solid**	**Dark grey / grey-black** (purple vapour when heated)	114	184
Astatine	At_2	Solid (predicted)	Black (predicted)	~300 (est.)	~340 (est.)

Chlorine, bromine and iodine at room temperature



The pattern in physical properties

Going down Group 7:

- The **colour gets darker**.
- **Melting and boiling points increase**, so the state changes from gas → liquid → solid. This is because the molecules get larger, so the **intermolecular forces** between them become stronger and more energy is needed to overcome them.

Worked Example: Use the pattern to predict the state and colour of fluorine and astatine at room temperature.

Fluorine is above chlorine: lower melting and boiling points than chlorine, so it is a **gas**; paler colour, so **pale yellow**.
Astatine is below iodine: higher melting point than iodine (114 °C), so a **solid**; darker than iodine, so **black**.

The test for chlorine

Test for chlorine gas: hold **damp blue litmus paper** in the gas. Chlorine turns the litmus paper **red** and then **bleaches it white**.

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The paper must be **damp** because chlorine reacts with water to form acids, one of which is a bleach. Chlorine is toxic, so this test should be carried out in a fume cupboard.

> **Exam Tip:** Write "**damp** litmus paper is **bleached**" — both words matter. Saying it "turns white" without "bleached" may not score, and dry litmus paper does not work.

Reactions with metals — forming metal halides

Halogens react with metals to form **metal halides**, which are ionic salts containing halide ions (F^-, Cl^-, Br^-, I^-):

metal + halogen → metal halide

$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$ (sodium burns with a bright orange flame; white solid forms)

$2Fe(s) + 3Cl_2(g) \rightarrow 2FeCl_3(s)$ (hot iron wool glows; brown solid forms)

$Mg(s) + Br_2(g) \rightarrow MgBr_2(s)$

The **reactivity decreases down the group**: chlorine reacts most vigorously with hot iron wool, bromine less vigorously and iodine only slowly. Following the pattern, fluorine reacts explosively with most metals even when cold, and astatine would react very slowly.

Halogen with hot iron wool	Observation	Product
Chlorine	Iron wool glows brightly	Iron(III) chloride (orange-brown solid)
Bromine vapour	Iron wool glows, less brightly	Iron(III) bromide (red-brown solid)
Iodine vapour	Iron wool glows only faintly; slow	Iron(II) iodide (grey solid)

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Reactions with hydrogen — hydrogen halides

Halogens react with **hydrogen** to form **hydrogen halides**, which are colourless gases:

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g) \quad \text{(hydrogen chloride)}$$

$$H_2(g) + Br_2(g) \rightarrow 2HBr(g) \quad \text{(hydrogen bromide)}$$

Hydrogen halides **dissolve in water** to form **acidic solutions**: hydrogen chloride forms **hydrochloric acid**, HCl(aq), which turns blue litmus red. Again reactivity decreases down the group: hydrogen and fluorine explode even in the dark and cold; hydrogen and chlorine explode in sunlight; hydrogen and iodine react only slowly and reversibly when heated. Astatine would form hydrogen astatide, HAt, which would also dissolve to give an acidic solution.

Worked Example: Predict the product when hydrogen reacts with iodine, and what forms when the product dissolves in water.

Pattern: $H_2 + X_2 \rightarrow 2HX$. So **$H_2(g) + I_2(g) \rightarrow 2HI(g)$** — hydrogen iodide (reaction slower than for chlorine or bromine).
Hydrogen iodide dissolves in water to form an **acidic solution** (hydroiodic acid), pH below 7.

6.4 Halogen Displacement Reactions

A **more reactive halogen** will **displace** a **less reactive halogen** from an aqueous solution of its halide (salt). The order of reactivity is **chlorine > bromine > iodine**.

chlorine + potassium bromide → potassium chloride + bromine

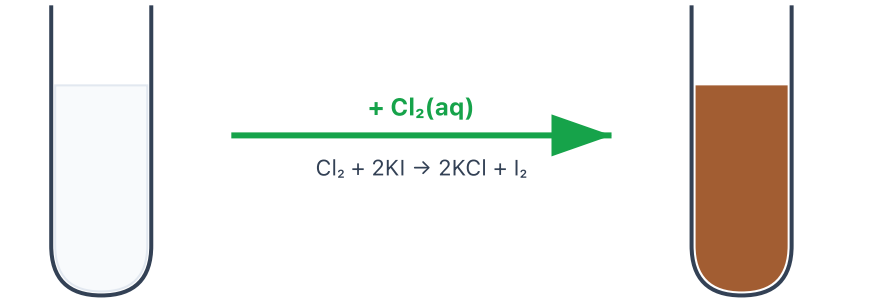
$$Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)$$

Halide solutions (KCl, KBr, KI) are all colourless, so a colour change shows that a reaction has happened and the free halogen has been released.

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Halogen water added ↓ / halide solution →	Potassium chloride, KCl	Potassium bromide, KBr	Potassium iodide, KI
Chlorine water (very pale green)	—	✓ Turns **orange / yellow-orange** (Br_2 formed)	✓ Turns **brown** (I_2 formed)
Bromine water (orange)	✕ No reaction (stays orange)	—	✓ Turns **brown** (I_2 formed)
Iodine solution (brown)	✕ No reaction	✕ No reaction	—

Chlorine water added to potassium iodide solution



KI(aq): colourless — brown: iodine released

Chlorine is more reactive than iodine, so it takes iodine's place in the salt.

Carrying out halogen displacement reactions

Although not a core practical, Edexcel often asks you to describe or interpret this experiment:

1. Place about 2 cm^3 of each halide solution (potassium chloride, potassium bromide, potassium iodide) in separate test tubes, or in the wells of a spotting tile.
2. Add a few drops of **chlorine water** to each and look for a colour change against a white background.
3. Repeat with fresh halide solutions using **bromine water**, then **iodine solution**.
4. Record “reaction / no reaction” and the colour in a results grid like the one above.

Safety: chlorine and bromine are toxic and corrosive — use dilute solutions, work in a well-ventilated room or fume cupboard and wear eye protection. **Variables:** keep the volume and concentration of each solution the same so the colour changes can be compared fairly.

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Exam Tip: Bromine water and iodine solution are both orange-brown at some concentrations, so a colour change from orange to brown can be hard to judge. Examiners accept "orange/yellow" for bromine and "brown" for iodine in water. If you are given a *different* colour description in the question, use the question's wording.

Predicting the reactions of astatine

Astatine is at the bottom of Group 7, so it is the **least reactive** of these halogens. It would be displaced by all of chlorine, bromine and iodine from astatide solutions, and astatine would **not** displace any of them:

$$I_2(aq) + 2NaAt(aq) \rightarrow 2NaI(aq) + At_2$$

$$At_2 + 2NaI(aq) \rightarrow \text{no reaction}$$

Displacement as redox (HT)

Halogen displacement reactions are **redox** reactions. The more reactive halogen **gains electrons** (is **reduced**) to form halide ions; the halide ions of the less reactive halogen **lose electrons** (are **oxidised**) to form the halogen. The metal ions (K^+, Na^+) are **spectator ions**.

$$\text{Ionic equation: } Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$$

$$\text{Reduction: } Cl_2 + 2e^- \rightarrow 2Cl^- \quad | \quad \text{Oxidation: } 2Br^- \rightarrow Br_2 + 2e^-$$

Worked Example: (HT) Bromine water is added to sodium iodide solution. Write the ionic equation and half equations, and state which species is oxidised and which is reduced.

Full equation: $Br_2(aq) + 2NaI(aq) \rightarrow 2NaBr(aq) + I_2(aq)$
Na^+ ions are spectators. Ionic equation: **$Br_2(aq) + 2I^-(aq) \rightarrow 2Br^-(aq) + I_2(aq)$**
$Br_2 + 2e^- \rightarrow 2Br^-$ (gain of electrons → **bromine is reduced**)
$2I^- \rightarrow I_2 + 2e^-$ (loss of electrons → **iodide ions are oxidised**)

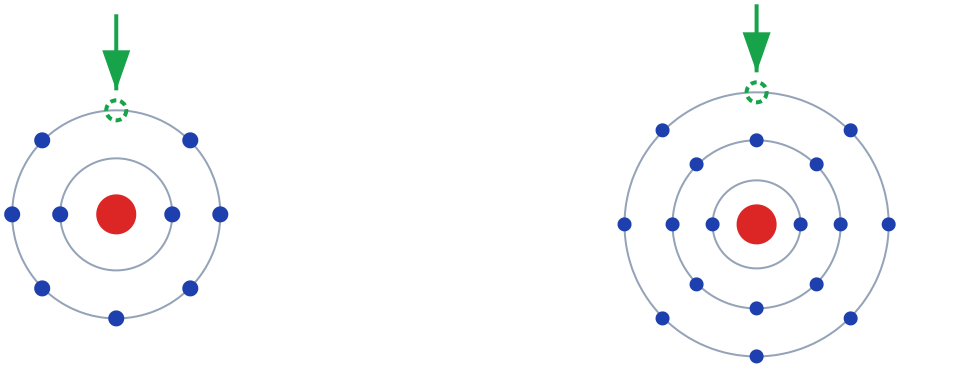
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Exam Tip: Be precise: it is the **iodide ions** that are oxidised, not "iodine". And bromine **molecules** are reduced to bromide **ions**. Remember OIL RIG, and check that electrons appear on opposite sides of the two half equations.

Explaining the reactivity trend in terms of electronic configuration

Halogen atoms have **7 outer electrons**. When they react they **gain one electron** to complete their outer shell and form a 1– ion (e.g. Cl + e^- → Cl^-; for the molecule, $Cl_2 + 2e^- \rightarrow 2Cl^-$).

Attracting an extra electron: fluorine vs chlorine



F 2.7
outer shell close to nucleus

Cl 2.8.7
outer shell further away, more shielding

Going down the group:

1. The atoms have **more electron shells** and are **larger**.
2. The outer shell is **further from the nucleus** (and more shielded by inner electrons).
3. So there is a **weaker attraction** between the nucleus and an incoming electron.
4. It is **harder to gain** an electron, so the halogen is **less reactive**.

Exam Tip: The halogen explanation is the **mirror image** of the Group 1 explanation. Group 1: reactivity increases down the group because the outer electron is *lost* more easily. Group 7: reactivity decreases down the group because an electron is *gained* less easily. Make sure you say "gain" for halogens — writing "lose" is a common, costly slip.

	Group 1 (alkali metals)	Group 7 (halogens)

Outer electrons	1	7
On reacting	Lose 1 electron → 1+ ion	Gain 1 electron → 1– ion
Reactivity down the group	**Increases**	**Decreases**
Melting point down the group	Decreases	Increases
Reason	Outer electron further from nucleus → weaker attraction → **easier to lose**	Outer shell further from nucleus → weaker attraction for incoming electron → **harder to gain**

6.5 Group 0 — The Noble Gases

Why the noble gases are inert

> The noble gases are **chemically inert** (very unreactive) because their atoms have a **full outer shell** of electrons (2 for helium, 8 for the others). Their electronic configuration is already **stable**, so they have no tendency to **lose, gain or share** electrons. As a result they exist as **single atoms** (monatomic gases) rather than molecules.

Element	Electronic configuration
Helium, He	2
Neon, Ne	2.8
Argon, Ar	2.8.8

Uses of the noble gases

The uses of noble gases depend on their **inertness**, **low density** and/or **non-flammability**:

Noble gas	Use	Property the use depends on
Helium	Filling party balloons and airships	**Low density** (much less dense than air, so it floats) and **non-flammable** (safer than hydrogen, which is also less dense but explosive)
Argon	Filling filament light bulbs	**Inert** — the very hot tungsten filament does not react (it would burn in air/oxygen)
Argon	Shielding gas for welding	**Inert** — prevents the hot metal from reacting with oxygen in the air
Argon	Double glazing (filling the gap between panes)	Inert, and a poor conductor of heat
Neon	Advertising signs (glows red when a current passes)	Inert — does not react with the electrodes when electricity passes through
Krypton	Some lasers and high-performance bulbs	Inert

> **Exam Tip:** For "why is helium used in balloons and not hydrogen?" you need **both** properties: helium has a low density (so the balloon rises) **and** it is non-flammable (hydrogen is flammable/explosive). One property alone gets only one mark.

Patterns in physical properties

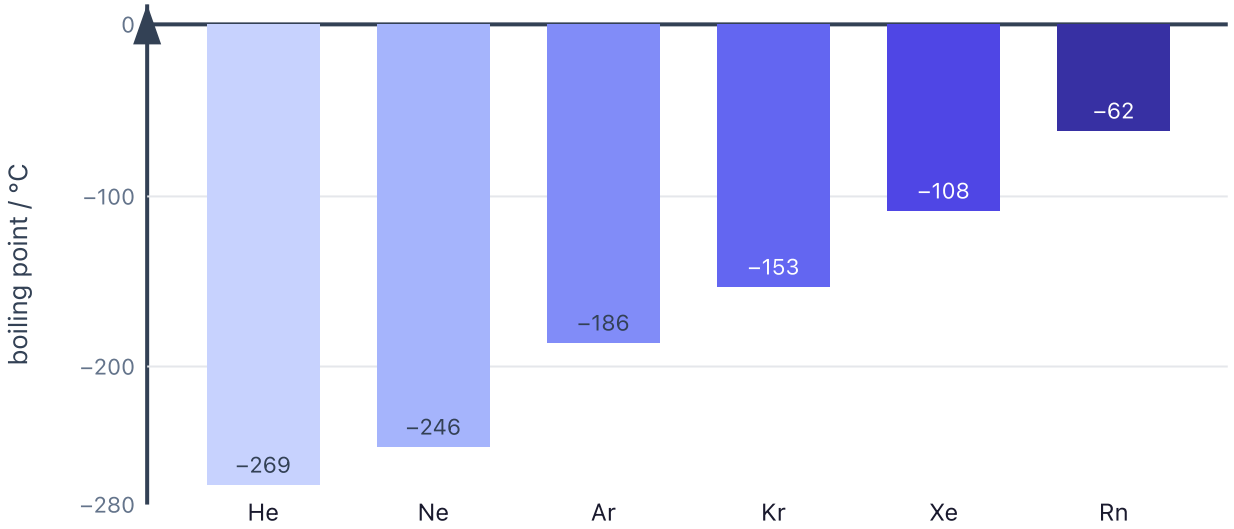
Going down Group 0:

- **Boiling point increases** (the atoms get larger and the forces between atoms get stronger).
- **Density increases** (the atoms get heavier).
- All are colourless gases at room temperature.

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Element	Relative atomic mass	Boiling point / °C	Density at RTP / g dm^{-3}
Helium	4	−269	0.17
Neon	20	−246	0.84
Argon	40	−186	1.66
Krypton	84	−153	3.48
Xenon	131	−108	5.46
Radon	222	−62	9.07

Boiling points of the noble gases



Bars closer to 0 °C = higher boiling point. Boiling point rises down the group.

Worked Example: The boiling points of argon and xenon are −186 °C and −108 °C. Predict the boiling point of krypton.

Krypton lies between argon and xenon, so its boiling point should be roughly halfway: (−186 + −108) ÷ 2 = **about −147 °C** (actual value −153 °C). Any answer between −186 and −108 °C that follows the trend is accepted.

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Worked Example: Radon has a density of about 9 g dm^{-3} and air about 1.2 g dm^{-3}. Explain why radon gas can build up in the basements of houses.

Radon is **much denser** than air (following the trend that density increases down Group 0), so it **sinks** and collects in low-lying spaces such as basements.

6.6 Using Group Trends to Make Predictions

Many Topic 6 questions give you data for some elements in a group and ask you to predict a property of another. Use these rules:

- Identify the **direction of the trend** (increasing or decreasing down the group) from the data given.
- Place the unknown element in the correct position and make sure your prediction **continues the trend** — for an element in the middle, give a value **between** its neighbours.
- For chemical reactions, predict the **same type of product** as the other members (hydroxide + hydrogen for Group 1; metal halide or hydrogen halide for Group 7) and a **faster or slower** reaction according to the reactivity trend.
- Give a **reason** that refers to the position in the group ("because it is lower down Group 1, so…").

Group	Property	Trend down the group	Prediction for an element below the last one shown
1	Reactivity with water	Increases	More vigorous reaction; more likely to explode
1	Melting point	Decreases	Lower melting point (caesium melts on a warm day, 28 °C)
7	Reactivity	Decreases	Slower reactions with metals and hydrogen; displaced by the halogens above it

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7	Melting/boiling point, colour	Increase; darker	Higher melting point, more likely solid, darker colour
0	Boiling point, density	Increase	Higher boiling point and higher density
0	Reactivity	All very unreactive	Still inert (full outer shell)

Worked Example: Francium is the element below caesium in Group 1. Predict (a) its electronic arrangement in terms of outer electrons, (b) how it would react with water, (c) the formula of its chloride.

(a) Like every Group 1 element it has **one electron in its outer shell** (with more shells than caesium).
(b) It would react **extremely violently / explosively** with water, forming francium hydroxide, FrOH (an alkaline solution), and hydrogen — because its outer electron is furthest from the nucleus and is lost most easily.
(c) It forms a 1+ ion, Fr^+, and chloride is Cl^-, so the chloride is **FrCl**.

Worked Example: Sodium reacts with chlorine to form sodium chloride. Explain, in terms of electrons, what happens to each atom.

Sodium (2.8.1) **loses** its one outer electron to form Na^+ (2.8). Chlorine (2.8.7) **gains** that electron to form Cl^- (2.8.8). Both ions now have a full outer shell — the same stable electronic configuration as a noble gas (neon and argon respectively). The oppositely charged ions attract to form ionic sodium chloride: $2Na + Cl_2 \rightarrow 2NaCl$.

Key Points Summary

- **6.1** — Classify alkali metals (Group 1), halogens (Group 7) and noble gases (Group 0) by their position and outer-shell electrons.

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- **6.2** — Alkali metals are soft with relatively low melting points.
- **6.3** — Li, Na and K react with water: float, fizz, form metal hydroxide (alkaline) + hydrogen; K ignites with a lilac flame. $2M + 2H_2O \rightarrow 2MOH + H_2$.
- **6.4** — Reactivity increases down Group 1; predict that Rb and Cs react even more violently.
- **6.5** — Explain: more shells → outer electron further from nucleus → weaker attraction → lost more easily.
- **6.6, 6.7** — Cl_2 pale yellow-green gas; Br_2 red-brown liquid; I_2 grey-black solid. Down Group 7: darker colour, higher melting/boiling points; predict F and At.
- **6.8** — Test for chlorine: damp blue litmus paper turns red then is bleached.
- **6.9, 6.10** — Halogens + metals → metal halides; halogens + hydrogen → hydrogen halides, which dissolve in water to form acids; reactivity decreases down the group.
- **6.11** — A more reactive halogen displaces a less reactive one from halide solutions ($Cl_2 > Br_2 > I_2$); predict that astatine is displaced by all of them.
- **6.12 (HT)** — Halogen displacement is redox: the halogen gains electrons (reduced), the halide ion loses electrons (oxidised); write ionic and half equations.
- **6.13** — Explain: more shells → outer shell further from nucleus → weaker attraction for an extra electron → harder to gain → less reactive.
- **6.14** — Noble gases are inert because they have full outer shells (stable electronic configuration).
- **6.15** — Uses depend on inertness (argon in bulbs and welding), low density and non-flammability (helium balloons).
- **6.16** — Boiling point and density increase down Group 0; use the trend to predict values for other noble gases.

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